

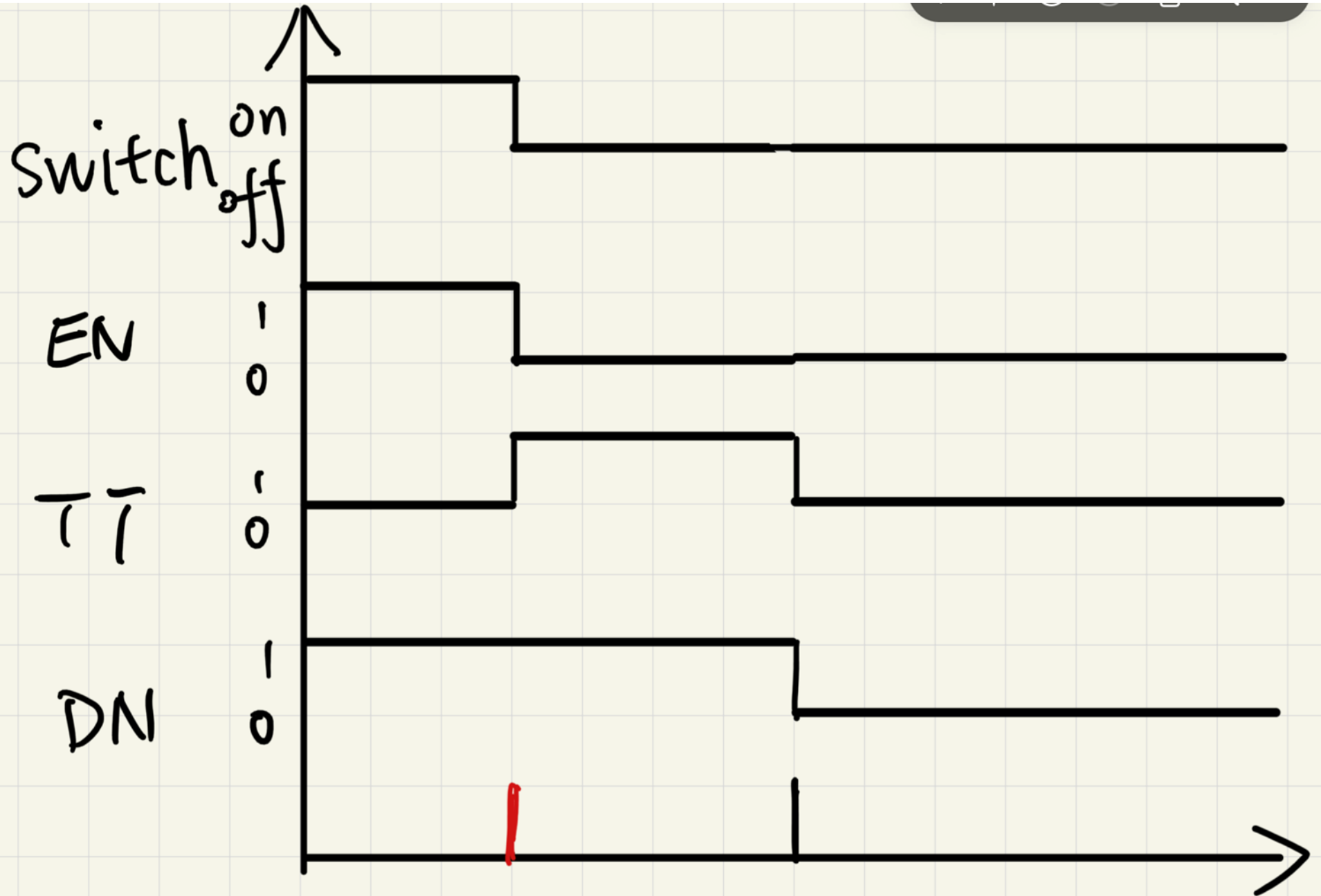
MMET 410 – Lab 5

- **Set up RSLogix 5000 software modules**
- **Have the worksheet pulled out on the side!

Instructions

- You need to include task 1,2,3,4,5 in your report
- For tasks 4, and 5, the format will be the same as previous labs
 - description of the program in own words
 - your input and output address
 - program screenshot or picture
- The deadline for report submission is tomorrow midnight

TOF
Timer
Off
Delay



Task 1/2/3

- **Answer all questions by yourself.** If you are uncertain about how TON, TOF, or RTO works, you could refer the links below
http://people.tamu.edu/~hsieh/hsieh/Hsieh_VirtualPLC.html
<https://twcontrols.com/lessons/rslogix-500-training-timers-ton-tof-and-rto>

Task 4

- Switch off -> Switch on -> switch off-> air_blast turn on for 2s -> air_blast turn off
- One TOF timer, one input (**switch**) and one output (**air_blast**)
- Use TT bit of the TOF
- Use **IN-4** for **switch**, and **OUT-0** for **air_blast**

Task 4: Inputs and Outputs

Type	Tag	Address / Preset	Function
Input	switch	Local:8:I.Data.4 (IN-4)	Part sensor (ON = part present)
Output	air_blast	Local:9:O.Data.0 (OUT-0)	Air blast valve
Timer	timer_1 (TOF)	2000 ms	2 s blast after part leaves

Task 5A

- switch on -> oil pump on -> (10s) -> shaft on -> oil pump still on
- switch **off** -> shaft **off** -> (15s) -> oil pump off
- Two timer (TON and TOF)
- Use TON to delay the ON signal
- Use TOF to delay the OFF signal
- Input : **Switch** IN-4
- Output : **shaft** OUT-0 / **oil_pump** OUT-1

Task 5A: Inputs and Outputs

Type	Tag	Address / Preset	Function
Input	switch (sw0)	Local:8:I.Data.4 (IN-4)	ON = run, OFF = stop
Output	shaft	Local:9:O.Data.0 (OUT-0)	Shaft motor
Output	oil_pump	Local:9:O.Data.1 (OUT-1)	Oil pump motor

Task 5B

- switch on -> oil pump on -> (10s) -> shaft on -> (oil runs for 180s) -> shaft off (and) LT3 on -> push reset -> reset RTO
- you need **one more** timer (RTO)
- **two more** outputs, **LT3** OUT-2 and a **RES** for RTO
- one more input, a **reset button** (IN-0)

Task 5B: Inputs and Outputs

Type	Tag	Address / Preset	Function
Input	switch (sw0)	Local:8:I.Data.4 (IN-4)	ON = run, OFF = stop
Input	PB4	Local:8:I.Data.0 (IN-0)	Resets timer3 (RES)
Output	shaft	Local:9:O.Data.0 (OUT-0)	Shaft motor
Output	oil_pump	Local:9:O.Data.1 (OUT-1)	Oil pump motor
Output	LT3	Local:9:O.Data.2 (OUT-2)	Change filter and oil

Task 4

- One TOF timer, one input (**switch**) and one output (**air_blast**)
- Use TT bit of TOF
- Use **IN-4** for **switch**, and **OUT-0** for **air_blast**
- Switch off -> Switch on -> switch off -> **air_blast** turn on for **2s** -> **air_blast** turn off

Task 5A

- Two timer (TON and TOF)
- Use TON to delay the ON signal
- Use TOF to delay the OFF signal
- Input : **Switch** IN-4
- Output : **shaft** OUT-0 / **oil_pump** OUT-1
- switch on -> oil pump on -> (**3s**) -> shaft on -> oil pump still on
- switch **off** -> shaft **off** -> (**5s**) -> oil pump off

Task 5B

- **one more** timer (RTO)
- **two more** outputs, **LT3** OUT-2 and a **RES** for RTO
- one more input, a **reset button** (IN-0)
- switch on -> oil pump on -> (**3s**) -> shaft on -> (oil for **20s**) -> shaft off (and) **LT3** on -> push reset -> reset RTO